Scalars & Vectors

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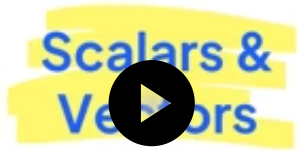

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Scalar & vector quantities

Extended tier only

- All quantities can be one of two types:
  - A **scalar**
  - A **vector**

Scalars

- Scalar quantities have only a **magnitude**
  - **Mass** is an example of a scalar quantity because it has magnitude **without direction**
  - **Energy** and **volume** are also examples of scalar quantities

Vectors

- Vector quantities have both **magnitude** and **direction**
  - **Weight** is an example of a vector quantity because it is a **force** and therefore has both **magnitude** and **direction**
  - **Acceleration** and **momentum** are also examples of vector quantities

Distance and displacement

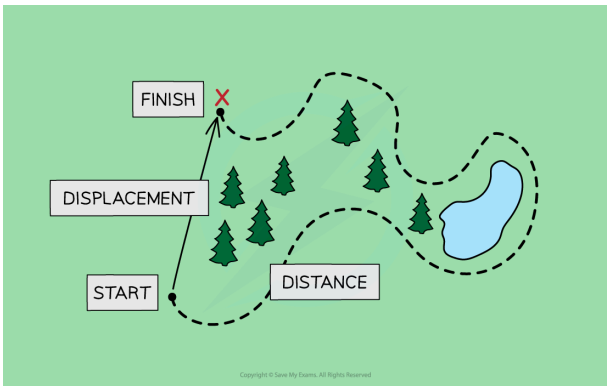
- **Distance** is a measure of how far an object has travelled, **regardless of direction**
  - Distance is the total length of the path taken

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- **Displacement** is a measure of how far it is between two points in space, **including the direction**
  - Displacement is the **length** and **direction** of a **straight line** drawn from the starting point to the finishing point
  - Displacement, therefore, has a **magnitude** and a **direction**
  - So, displacement is a **vector** quantity

What is the difference between distance and displacement?



Displacement is a vector quantity, while distance is a scalar quantity

- When a student travels to school, there will probably be a **difference** between the **distance** they travel and their **displacement**
  - The **overall distance** they travel includes the total lengths of all the roads, including any twists and turns
  - The **overall displacement** of the student would be a straight line between their home and school, regardless of any obstacles, such as buildings, lakes or motorways, along the way

Speed and velocity

- **Speed** is a measure of the **distance** travelled by an object per unit time, **regardless of the direction**
  - The speed of an object describes how fast it is moving, but **not** the direction it is travelling in
  - Speed, therefore, has **magnitude** but **no direction**
  - So, speed is a **scalar** quantity

- **Velocity** is a measure of the **displacement** of an object per unit time, **including the direction**
  - The velocity of an object describes how fast it is moving **and** which direction it is travelling in
  - An object can have a **constant speed** but a **changing velocity** if the object is **changing direction**
  - Velocity, therefore, has **magnitude and direction**
  - So, velocity is a **vector** quantity

vectors

Extended tier only

- The table below lists some common examples of scalar and vector quantities
- Corresponding scalars and their vector counterparts are aligned in the table where applicable

Table of scalars and vectors

Scalar	Vector
distance	displacement
speed	velocity
mass	weight
	force
	acceleration
	momentum
	electric field strength
energy	
volume	
density	
temperature	
power	

Worked Example

An instructor is in charge of training junior astronauts. For one of their sessions, they would like to explain the difference between mass and weight.

Suggest how the instructor should explain the difference between mass and weight, using definitions of scalars and vectors in your answer.

Answer:

Step 1: Recall the definitions of a scalar and vector quantity

- Scalars are quantities that have only a magnitude

Step 2: Identify which quantity has magnitude only

- Mass is a quantity with magnitude only
- So mass is a scalar quantity
- The instructor might explain to their junior astronauts that their mass will not change as their location in the Universe changes

Step 3: Identify which quantity has magnitude and direction

- Weight is a quantity with magnitude and direction (it is a force)
- So weight is a vector quantity
- The instructor might explain that their weight, the force on them due to gravitational field strength, will vary depending on their location. For example, the force of weight acting on them would be less on the Moon than it is on Earth

Examiner Tips and Tricks

Make sure you are comfortable with the differences between **similar** scalars and vectors.

The most commonly confused pairings tend to be:

- distance and displacement
- speed and velocity
- weight and mass

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